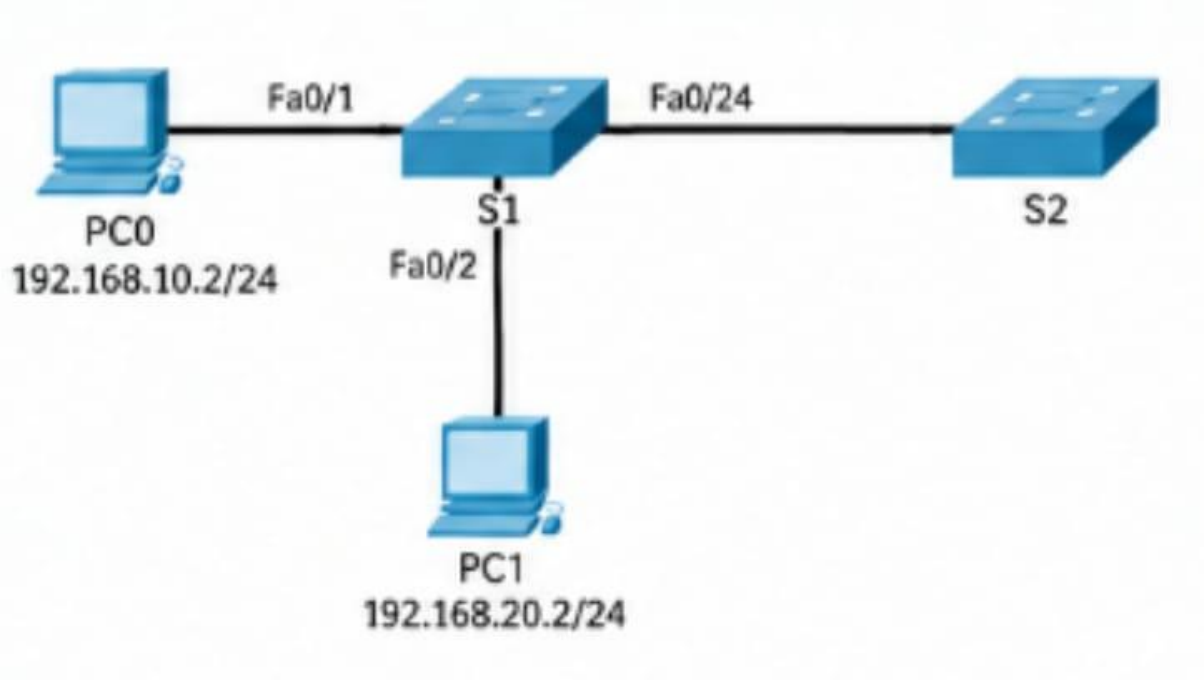
**EXPERIMENT 8**

**CONFIGURE ETHERNET SWITCH PORTS (ACCESS/TRUNK), AUTO/DUPLEX SETTINGS; OBSERVE LINK NEGOTIATION**

**NETWORK TOPOLOGY :**




**S1 CONFIGURATION ( ACCESS PORT – Fa0/1) :**

```
S1> enable
S1# configure terminal
S1(config)# vlan 10
S1(config-vlan)# name STUDENTS
S1(config-vlan)# exit
S1(config)# interface fastEthernet 0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# speed auto
S1(config-if)# duplex auto
S1(config-if)# no shutdown
S1(config-if)# end
S1#
```

## S1 – VLAN BRIEF & ACESS PORT STATUS :

```
S1# show vlan brief

VLAN  Name                             Status    Ports
----  -------------------------------- --------- -------------------------------
1     default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                 Fa0/7, Fa0/8, Fa0/9, Fa0/10
10    STUDENTS                         active    Fa0/1
20    STAFF                            active
1002  fddi-default                     active
1003  token-ring-default               active
1004  fddinet-default                  active
1005  trnet-default                    active

S1# show interfaces fastEthernet 0/1 switchport
 Name: Fa0/1
 Switchport: Enabled
 Administrative Mode: access
 Operational Mode: access
 Access Mode VLAN: 10 (STUDENTS)
 Trunking Native Mode VLAN: 1 (default)
 Voice VLAN: none
S1#
```

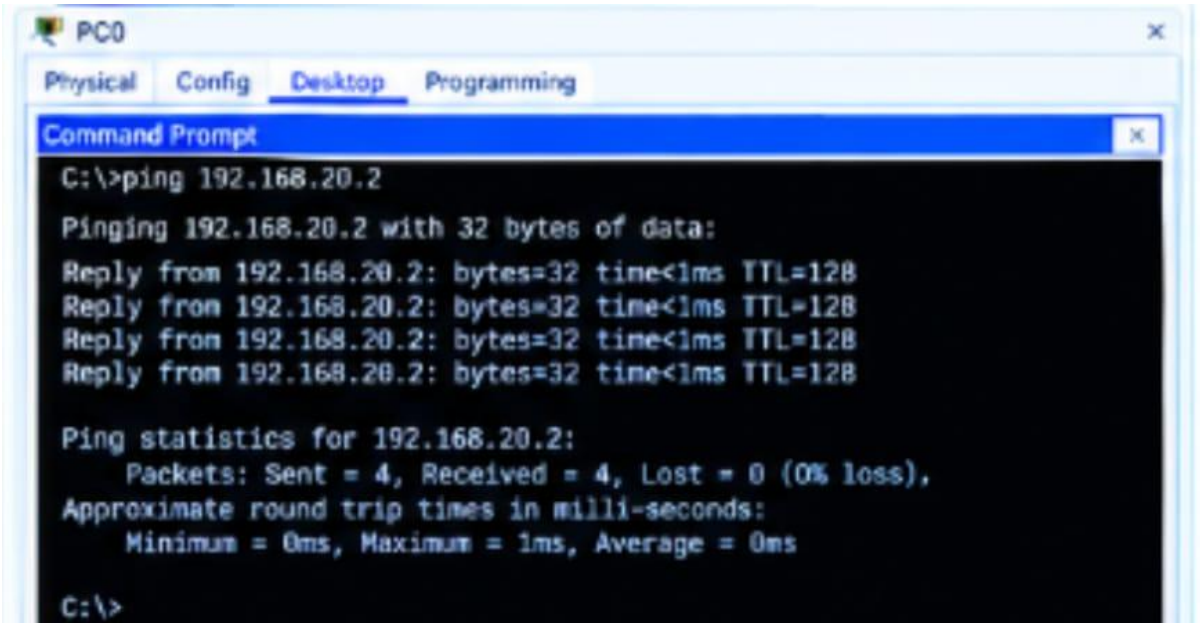
## S1 – TRUNK PORT CONFIGURATION ( Fa0/24 ):

```
S1(config)# interface fastEthernet 0/24
S1(config-if)# switchport mode trunk
S1(config-if)# no shutdown
S1(config-if)# end

S1# show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1

S1# show interfaces fastEthernet 0/24 switchport
 Name: Fa0/24
  Switchport: Enabled
  Administrative Mode: trunk
  Operational Mode: trunk
  Access Mode VLAN: 1 (default)
  Trunking Native Mode VLAN: 1 (default)
  Trunking VLANs Enabled: 1-4094
  Pruning VLANs Enabled: 2-1001,1006-1009
S1#
```

**PING TEST :**


PC0
Physical
Config
Desktop
Programming
Command Prompt
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>